The spherical accumulation kernel and the source-to-field interface

Chapter III / Source-bound development
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Abstract. A retained energy account becomes a spatial field through a specified localization, coupling and receiving operator. This chapter develops the isolated Poisson kernel, its energy and force proofs, and two complete finite constructions: the actual retained eight-site source profile under a declared physical embedding, and an exactly solvable shell network with boundary and continuum errors. A static Chapter I field-energy example and an exact interacting-clock coupling criterion connect the construction to Chapters I and II. Source identities, physical assignments and computed results are stated separately. All earlier drafts remain preserved.

1. The next source question

Chapter I supplies an interacting finite source and continuum field. Chapter II distinguishes a present output, a predictive state, a timed history and an unread energetic complement. Chapter III asks how their physical energy is located and received in space. [1,2]

The spherical lift is a static receiving profile:

$$a(r) = \frac{2GM}{c^2 r}, \quad A_{\mathrm{tot}} = A_0 + a, \quad A_0 = \frac{1}{12\pi}.$$

The physical mass M, source radius b, receiving radius r and Chapter I coupling radius R have different roles. G and c carry the declared physical units. The floor is dimensionless.

This revision carries exact source operands into finite field calculations. The mass/localization and coupling assignments are recorded at each step.

2. What enters the field

For a positive source form B and record r=L^{T}y, the ambient minimum is

$$C(r) = r^T \mathcal{G}^+ r, \quad \mathcal{G} = L^T B^{-1} L,$$

provided r belongs to the range of the receiver Gram. An infeasible record has no such source; its constrained infimum is +infinity. Allowed-state restrictions can raise the minimum. A change from an occupied reference also retains its cross term. [2]

This is a cost for a specified receiver. To form a physical density, specify the energy operator, physical conversion, reference and spatial incidence. The completed Chapter II now develops these requirements in its source-to-space interface.

The source and receiver below are static. Pressure, motion, retardation and nonlinear self-coupling require their own physical laws.

3. The declared receiving law

For a supplied nonrelativistic mass density ρ, use

$$\Delta a = -\frac{8\pi G}{c^2}\rho, \qquad \Phi = -\frac{c^2}{2}a.$$

This is equivalent to $\Delta \Phi = 4\pi G \rho$. The field normalization imports the Newtonian coupling. The sign makes a positive for a positive isolated mass when a tends to zero at infinity.

A test body with Lagrangian $L = m|\dot{x}|^2/2 - m\Phi$ has acceleration

$$\mathbf{g} = -\nabla\Phi = \frac{c^2}{2}\nabla a.$$

Equality of passive gravitational and inertial mass is part of this test-body coupling. The scalar profile by itself does not select that equality.

For bounded compact density the solution is

$$a(x) = \frac{2G}{c^2}\int \frac{\rho(y)}{|x-y|}\, d^3y.$$

The locally integrable kernel obeys $\Delta(1/r) = -4\pi$ delta. Linearity therefore gives the response to a sum of prescribed sources. Global regularity and isolated boundary data establish uniqueness; an exterior subproblem needs its inner data too.

All three displayed formulas have a fixed energy and time convention. A later rescaling of a must also rescale its source equation and its receiving laws.

4. Flux and exterior uniqueness

In a spherical vacuum exterior,

$$\frac{1}{r^2}(r^2 a')' = 0, \qquad a = C_0 + \frac{C_1}{r}.$$

Decay fixes C_0, while the enclosing source flux fixes C_1:

$$M(<S) = -\frac{c^2}{8\pi G}\oint_S \nabla a \cdot d\mathbf{S}.$$

Thus $C_1 = 2GM/c^2$ and $g = -GM\,\hat{r}/r^2$. Decay alone leaves the amplitude free. A regular vacuum ball containing the origin cannot carry a nonzero 1/r term.

For a nonspherical source, total mass fixes only the monopole. Higher multipoles carry shape information while contributing zero net enclosing flux. The cube construction below makes the first surviving shape term explicit.

A finite outer wall fixes a different boundary problem. Its correction must be included before comparing potential or clock values to the isolated profile. The shell example supplies that correction exactly, including its energy cost.

The spatially constant floor contributes no source flux. This statement is compatible with a nontrivial floor in an independently specified nonlinear clock receiver.

5. Finite spherical sources

For a spherical source define its enclosed mass:

$$M(r) = 4\pi \int_0^r \rho(s) s^2 \, ds.$$

Regularity and the source equation give

$$a'(r) = -\frac{2GM(r)}{c^2 r^2},$$

$$a(r) = \frac{2G}{c^2}\left[\frac{M(r)}{r} + 4\pi \int_r^\infty \rho(s) s \, ds\right].$$

Exterior shells supply potential and zero interior acceleration. A thin shell of radius b and mass M has a=2GM/(c^2b) inside and 2GM/(c^2r) outside. Its derivative jump carries its surface mass. Values directly on a zero-thickness surface require a limiting convention.

For a uniform ball,

$$a(r) = \frac{GM}{c^2 b}(3 - r^2/b^2), \qquad r \le b,$$

$$a(r) = \frac{2GM}{c^2 r}, \qquad r \ge b.$$

Both a and a' match at the surface. At the center a=3GM/(c^2b) while g=0. Bounded compact density makes the central expression regular; finite mass alone does not. A central ρ proportional to r^{-2} is a useful finite-mass, divergent-potential example.

6. Mechanical and tidal receivers

Circular motion differentiates the potential; escape compares it with a destination:

$$v_{\mathrm{circ}}^2 = -\frac{c^2 r}{2} a'(r),$$

$$v_{\mathrm{esc}}^2 = c^2[a(r) - a(\infty)].$$

For the isolated spherical exterior they are GM/r and 2GM/r. A shell interior has zero circular-force field yet a nonzero escape energy relative to infinity. A truly global potential shift cancels when both endpoint values shift together.

For the exterior monopole, the tidal tensor is

$$T_{ij} = \frac{GM}{r^3}(3 n_i n_j - \delta_{ij}).$$

It has zero trace in vacuum, radial stretching 2GM/r^3, and transverse compression -GM/r^3. Inside a uniform ball it is -GM delta_ij/b^3, including at the zero-acceleration center; its trace is -4π G ρ.

The tidal tensor here differentiates Newtonian acceleration. A relativistic spatial curvature or photon receiver requires additional structure. Chapter I's phase clock is another receiving operation, whose source coupling is derived below.

7. Radial controls and scale

For a trial $a_p = K r^{-p}$ away from the origin,

$$\Delta a_p = p(p-1)Kr^{-p-2},$$

$$g_r = -\frac{c^2 pK}{2} r^{-p-1},$$

$$M_{\mathrm{flux}}(r) = \frac{c^2 pK}{2G} r^{1-p}.$$

A nonzero decaying spherical vacuum solution with constant source flux selects p=1. Positive attraction alone leaves many powers. The harmonic p=0 term is constant and carries no mass flux.

Compare powers using a reference radius: $a_p = a_* (r_*/r)^p$. Matching the potential at r_* gives a force p times the monopole force there. Dividing this profile by p matches that force but changes the potential to 1/p of its target. At other radii the force ratio is $(r_*/r)^{(p-1)}$.

The coefficient K has units length^p. Comparing its numerical value across powers without the reference radius changes the dimensional question. A matched datum at one radius therefore supplies less information than a radial profile, flux and potential reference together.

8. A finite-energy domain

For an isolated field, use the homogeneous Sobolev space $D^{1,2}(\mathbb{R}^3)$: the completion of compact smooth functions under $\|\nabla a\|_2$, represented in L^6. This fixes the infinity reference and excludes a nonzero constant ambiguity.

A sufficient source class is ρ in $L^{6/5}$; bounded compact density is an example. The pairing integral ρ a is then bounded by Holder and Sobolev. The functional

$$J[a;\rho] = \frac{c^4}{32\pi G}\int |\nabla a|^2$$

$$-\frac{c^2}{2}\int \rho a$$

has a unique weak minimizer. Its Euler equation is the declared Poisson problem. Existence follows from the bounded source functional and the positive gradient inner product; uniqueness follows from the same norm.

Compact smooth test variations give

$$\delta J = \int \left[-\frac{c^4}{16\pi G}\Delta a - \frac{c^2}{2}\rho \right] \delta a.$$

An ideal point source still has a distributional exterior solution, but has infinite self-energy. Thin spherical shells have a finite-energy potential with a derivative jump. The finite cube below uses regular balls to specify its self-energy.

9. The minimum and binding energy

Let a_* be the weak solution and let v be an allowed variation with the same boundary reference. Expand $J[a_*+v]$. The linear terms cancel by the weak Poisson equation, leaving the exact identity

$$J[a_* + v] - J[a_*] = \frac{c^4}{32\pi G}\int|\nabla v|^2.$$

This proves the global minimum on the stated domain. Equality requires v=0 in $D^{1,2}$. For a bounded Dirichlet region the same proof uses zero-trace variations.

Testing the weak equation against a_* gives

$$\int|\nabla a_*|^2 = \frac{8\pi G}{c^2}\int\rho a_*.$$

Consequently,

$$U = J[a_*] = -\frac{c^2}{4}\int\rho a_*$$

$$= \frac{1}{2}\int\rho\Phi = -\frac{c^4}{32\pi G}\int|\nabla a_*|^2.$$

The positive penalty in J and the negative on-shell binding energy are different parts of the same calculation. For a uniform ball, integration over the whole field, including its exterior, gives $U=-3GM^2/(5b)$. Keeping only the material interior loses part of this energy.

10. Two sources and force

Let a_i be the isolated response to ρ_i and U_i its self-energy. Symmetry of the Green kernel gives

$$U[\rho_1 + \rho_2] = U_1 + U_2$$

$$-G\iint\frac{\rho_1(x)\rho_2(y)}{|x-y|}\,d^3x\,d^3y.$$

Proof: expand $-c^2$ integral(ρ_1+ρ_2)(a_1+a_2)/4. The two cross terms coincide and combine to the displayed pair interaction. The factor one-half in the self-energy removes double counting.

For spherical bodies whose enclosing balls are disjoint (d>b1+b2), the cross term is -GM1 M2/d, with d their center separation. Differentiation at fixed body shapes gives attraction of magnitude $GM1\,M2/d^2$. This uses the same G, density reference and mass assigned to the field equation. The force follows from that common energy account.

For general disjoint compact bodies, differentiation under the integral gives the vector force directly. Positive separation bounds the differentiated kernel and justifies the step. Holding a source fixed while varying its field differs from translating that source and re-solving the field; this proof tracks the latter through the on-shell energy.

11. The floor in a clock receiver

A constant floor has zero gradient. It also cancels from an ordinary endpoint subtraction. For the nonlinear diagnostic N_raw(a)=sqrt(1-A0-a), source-free normalization instead gives

$$\frac{N_{\mathrm{raw}}(a)}{N_{\mathrm{raw}}(0)} = \sqrt{1 - \frac{a}{1 - A_0}}.$$

Its logarithmic slope at zero is -1/[2(1-A0)]. With A0=1/(12pi), the residual scale factor is approximately 1.027248616.

The locally declared source-only lapse is

$$N_{\mathrm{loc}}(a) = \sqrt{1 - a}, \quad N'_{\mathrm{loc}}(0) = -\frac{1}{2}.$$

It gives the desired leading endpoint coefficient when a retains its Poisson normalization. Defining a_tilde=a/(1-A0) changes the source amplitude too; that redefinition must be carried through every equation.

These are explicitly different receiver prescriptions. The six-lane interacting-clock test below asks which perturbation of the full source Hamiltonian actually produces the source-only coefficient. Clock normalization alone supplies no density or absolute energy reference.

12. The weak metric interface

Fix static isotropic weak-field coordinates and write

$$ds^2 = -[1 - a + O(a^2)]c^2 dt^2$$
$$+[1 + \gamma a + O(a^2)]d\mathbf{x}^2.$$

The stationary clock coefficient is one-half. A null path in these coordinates instead has

$$c\,dt = \left[1 + \frac{1+\gamma}{2}a + O(a^2)\right] d\ell.$$

The photon-road coefficient depends on the spatial response gamma. In conventional post-Newtonian notation gamma=1 gives coefficient one. These metric coefficients are additional physical input; static Poisson and an endpoint clock do not determine gamma. [7]

The coordinate convention also matters beyond first order. An exact square-root lapse must be tied to a specified time and radial coordinate before it is used as a nonlinear prediction.

Chapter IV can develop an actual endpoint receiver with reference and motion terms, and Chapter V a path receiver with spatial response. The following finite source constructions provide spatial inputs for both while retaining their separate receiving operations.

13. The actual spatial source

The retained Si29 construction supplies exact finite spatial effects. Its four-component Gram Q has diagonal (128,136,128,128). Its only nonzero off-diagonal entries are 8 on pairs 01, 03 and 23, with symmetric entries. Thus every original entry is retained. [3]

Write h=√Q. The orthonormal bright generator has zero diagonal blocks and h in both off-diagonal blocks. Its positive excitation account is

$$H_{\mathrm{gen}}^2 = \mathrm{diag}(Q, Q), \qquad H_0 = \mathrm{diag}(h, h).$$

Here H0=|H_{gen}|, and the two commute. The signed coherent generator and its positive excitation account have different roles. Replacing the signed spectrum by its magnitude is not a common scalar shift.

For species t at site x, let Ntx be the exact incoming occupation and Gtx the outgoing number Gram. Their retained denominator is 240. Define

$$O_{tx} = \frac{1}{8}\mathrm{diag}(N_{tx}, W_{tx}),$$

$$W_{tx} = \frac{1}{240} h^{-1} G_{tx} h^{-1},$$

$$E_{tx} = H_0^{1/2} O_{tx} H_0^{1/2}.$$

Every Ntx is nonnegative and every Gtx is a positive Gram. Congruence makes Etx positive. Summing every species and site returns H0. The actual source evolution, energy transfer, current and coherence retain their original definitions. [3]

This account is an inherited excitation-energy choice. It is not added to H_{gen} as another term or identified with complete rest energy. A spatial density also needs an embedding, physical energy conversion and a common reference.

14. An exact flatness identity

At each of the eight sites the retained matrices obey

$$\sum_t N_{tx} = I, \qquad \sum_t G_{tx} = 240Q.$$

Since Q=h², substitution into the effects gives the short proof

$$\sum_t O_{tx} = \frac{I}{8}, \qquad E_x := \sum_t E_{tx} = \frac{H_0}{8}.$$

Consequently Ex commutes with H_{gen}. Under H_{gen}, every preparation has a stationary one-eighth share of this excitation energy at each site, although species and coherence evolve in the exact 36-coordinate closure.

The total amplitude can still change with preparation. Retained evaluations are:

Preparation	Total relative energy
e0	11.302793352678457
plus	11.828266226372410
minus	11.131220607210203
phase	11.479743416791305

Each site's energy is the displayed total divided by eight. The fractions are exact; these decimals are retained evaluations. This scalar receiver fixes a spatial shape and does not retain every changing source coordinate.

The independent checker verifies all spatial Gram principal minors, all eight flatness identities, and agreement with the separately retained observable closure. Two native operator words receive the complete incoming and outgoing identities. Their actual execution is recorded in the campaign receipt map.

The next density question is therefore precise: justify this uniform physical localization or derive a different local energy/stress coupling from the source. Species weights cannot be selected merely to obtain a desired field.

15. A finite cube localization

Choose the orthogonal cube embedding Xx=ℓ times ξ_x, with every component of ξ_x equal to +1 or -1. It matches the retained cube-edge construction. The edge graph alone does not fix these angles. Its source-coordinate edge length is two; ℓ assigns a physical length.

Choose a normalized nonnegative spatial profile ζ_x and an energy unit eu. The explicit excitation-mass account is

$$\rho_{\mathrm{exc}}(z) = \frac{e_u}{c^2}\sum_x \langle E_x\rangle \zeta_x(z - X_x),$$

$$M = \frac{e_u\langle H_0\rangle}{c^2}, \qquad \rho_{\mathrm{exc}} = \frac{M}{8}\sum_x \zeta_x.$$

For the complete worked case, each profile is a uniform ball of radius δ<ℓ and volume 4π δ³/3. The balls are disjoint; each carries mass M/8. This specifies finite interiors and finite self-energy.

The embedding, smoothing radius and E/c² conversion are stated assignments. A compatible coupling uses the spatial average of a over each ball, not merely its center value. Other active fields, interactions and reference apparatus still need their own spatial account.

Under the declared Poisson receiver, a point outside every ball has

$$a(z) = \frac{2GM}{c^2}K_{\mathrm{cube}}(z),$$

$$K_{\mathrm{cube}}(z) = \frac{1}{8}\sum_x \frac{1}{|z - X_x|}.$$

The shell theorem makes this equality exact outside the balls. It does not make the cube spherical. The enclosing flux returns M, and the exterior profile retains shape information beyond its monopole.

Declare $H_{gen}(a)=H_{gen}-aH_0/2$ for a uniform lift. For the actual incoming/outgoing contrast B=diag(I,-I), H_0 commutes with B and H_{gen}, so its entire history is unchanged. Scaling H_{gen} itself gives time dilation. This is the Si29 population clock; Chapter I uses a different, certified phase receiver.

16. Shape and a controlled remainder

Let r=|z|, n=z/r and q=sqrt(3)ℓ/r<1. The cube centers have common radius sqrt(3)ℓ. Inversion symmetry removes odd multipoles. Its second moment is ℓ² times the identity, so the traceless quadrupole also vanishes.

The fourth moment is

$$\frac{1}{8}\sum_x (n\cdot X_x)^4 = \ell^4\left(3 - 2\sum_j n_j^4\right).$$

Insert this into P4(u)=(35u⁴-30u²+3)/8. The first surviving correction is

$$K_{\mathrm{cube}}(rn) = \frac{1}{r} + \frac{A_4(n)}{r^5} + R_6,$$

$$A_4(n) = \frac{7\ell^4}{4}\left(3 - 5\sum_j n_j^4\right).$$

Since |Pl|<=1, the remaining even Legendre series is bounded by

$$|R_6| \le \frac{q^6}{r(1-q^2)}.$$

The monopole-only bound is q⁴/[r(1-q²)]. The degree-four correction is -7ℓ⁴/(2r⁵) on a cube axis and +7ℓ⁴/(3r⁵) along a body diagonal.

This is a source-shape control with no amplitude fit. Its domain requires r>sqrt(3)ℓ and an observation point outside every ball. The smearing changes the interior and self account while leaving this exterior center sum unchanged.

Low-order isotropy and an exactly spherical field are different conclusions. The finite cube supplies the former through degree three. A proposed macroscopic isotropic limit must specify how source orientations, placements or a graph family approach that limit, and bound the surviving shape terms.

17. Finite cube binding energy

Eight uniform balls each have mass m=M/8 and radius δ. Their self energies sum to

$$U_{\text{self}} = 8\left(-\frac{3Gm^2}{5\delta}\right) = -\frac{3GM^2}{40\delta}.$$

The cube has 12 unordered pairs at distance 2ℓ, 12 at 2sqrt(2)ℓ, and four at 2sqrt(3)ℓ. Nonoverlap and spherical symmetry give each exact center-pair interaction. Thus

$$U_{\text{pair}} = -\frac{GM^2}{64\ell}\left(6 + 3\sqrt{2} + \frac{2}{\sqrt{3}}\right).$$

The sum is the on-solution Newtonian binding account for this chosen density, using the whole field and one infinity reference. The source masses in the linear equation remain the prescribed M/8. Adding U/c² back into that source would introduce a further self-gravity completion.

This calculation also identifies which energy is missing from an excitation-only mass assignment. A complete account must include active field, interaction, counterterm and reference/apparatus energies under the same reference. Bond interactions need not possess an arbitrary positive site decomposition.

In particular, a positive minimum record cost and this negative binding account are different quantities. Counting the on-solution binding energy and then adding its source-field interaction a second time would double count the same coupling. The finite localization makes this distinction calculable rather than leaving an unregularized point-source self-energy implicit.

18. An exact radial receiver

The retained shell conductance is

$$c_i = i(i+1) = \left(\int_i^{i+1} \frac{du}{u^2}\right)^{-1}.$$

Use radii ri=ih, an outer boundary aL=0, and no inner sink, c0=0. Source shells have masses Munit mi. With kappa=2GMunit/(c²h), solve

$$Ka = \kappa m, \qquad K = B^T C B,$$

where B takes outward edge differences and C=diag(ci). Its ith row is

$$(Ka)_i = c_i(a_i - a_{i+1}) + c_{i-1}(a_i - a_{i-1}).$$

For any nonzero v, vᵀKv is the positive sum of weighted squared edge differences. They cannot all vanish while the outer value is zero. K is therefore positive definite and the finite problem has a unique solution.

The outer exit carries an essential source flux. Removing it gives the closed graph Kclosed with Kclosed times 1 equal to zero. Summing its equations would require zero total injection. It cannot support this positive-total static source without an additional sink.

This is the retained spherical receiving geometry, with an explicit extended source and boundary. It is not an identification of concentric shells with Si29's eight cube sites. The two constructions answer distinct spatial questions while sharing a declared energy and receiver convention. [3]

19. Green kernel and flux proof

Sum the source equations through shell i. Interior edge fluxes cancel, leaving

$$c_i(a_i - a_{i+1}) = \kappa M_i, \qquad M_i = \sum_{j \le i} m_j.$$

The edge resistance telescopes:

$$\frac{1}{c_i} = \frac{1}{i} - \frac{1}{i+1}.$$

Summing the remaining differences to the outer boundary gives

$$H_{ij} = \frac{1}{\max(i,j)} - \frac{1}{L},$$

$$a_i = \kappa \sum_j H_{ij} m_j, \qquad KH = I.$$

No desired field values enter the right-hand side. The source and conductance determine them. Between shells use exact harmonic interpolation. On ri<r<ri+1 it gives

$$a'(r) = -\frac{2GM_{\text{unit}}M_i}{c^2 r^2},$$

$$g_r = -\frac{GM_{\text{unit}}M_i}{r^2}.$$

The flux returns the same enclosed mass. Each shell produces its derivative jump and a constant interior potential. At the shell itself choose a one-sided acceleration or the mean self-field used in radial virtual work.

This network samples the exact continuum thin-shell solution. The finite outer reference differs from isolation by the known constant 2GMtotal/(c²Lh). Gradients and fluxes agree; potential values and escape to infinity require that correction. Refining the radial mesh cannot remove a boundary-reference error held fixed in physical units.

20. A complete rational example

Take L=9, kappa=1/50, and the shell masses shown below. Their total is one. These fractions specify a test source, not an inferred microscopic mass spectrum.

i	mi	ai	Enclosed Mi
1	0	1777/630000	0
2	1/5	1777/630000	1/5
3	0	1357/630000	1/5
4	3/10	1147/630000	1/2
5	1/10	52/39375	3/5
6	0	29/31500	3/5
7	2/5	1/1575	1
8	0	1/3600	1

Every outgoing flux is kappa Mi, including the final exit. Let u=a/kappa and measure energy in GMunit²/h. The finite functional is F=u^{T}Ku/2-m^{T}u. Its minimum is -m^{T}Hm/2, giving

$$\frac{U_L}{GM_{\text{unit}}^2/h} = -\frac{9427}{252000},$$

$$\frac{U_\infty}{GM_{\text{unit}}^2/h} = -\frac{2603}{28000}.$$

The difference is the exact correction -GMtotal²/(2Lh). The exterior positive gradient term and the source-reference change combine into this signed on-solution result.

Varying a shell radius at fixed ordering gives

$$-\frac{dU}{dr_j} = -\frac{GM_j}{r_j^2}\left(M_{<j} + \frac{M_j}{2}\right).$$

The half self term is the mean of its two one-sided self-fields. This is radial work under spherical expansion, rather than net vector force on the shell center.

21. A controlled continuum limit

For a uniform ball of radius b, divide radius into n cells with h=b/n. Move each cell's mass to its outer endpoint:

$$\frac{m_j}{M} = \frac{j^3 - (j-1)^3}{n^3}.$$

Write eta=GM/(c²b), alpha=a/eta. The continuous profile is 3-x² inside and 2/x outside. The exact potential error obeys

$$0 \le \alpha_{\text{ball}} - \alpha_{\text{shells}} \le \frac{3}{n} - \frac{2H_n}{n^2},$$

where Hn is the nth harmonic sum. The bound is attained at the center. Moving a mass element from s to its endpoint sj reduces its kernel by at most 1/s-1/sj. Integrating that difference gives the displayed bound.

For binding energy the corresponding bound is

$$0 \le \frac{U_{\text{shells}}}{GM^2/b} + \frac{3}{5} \le \frac{3}{2n} - \frac{H_n}{n^2}.$$

Use 1/max(s,t)=min(1/s,1/t) in the pair integral; outward motion changes that minimum by at most the sum of the two reciprocal-radius changes. The energy factor one-half yields the bound.

A finite absorbing radius R adds reduced lift offset 2b/R and energy offset GM²/(2R). Isolation therefore requires both h/b→0 and b/R→0. Refining while R/b=2 leaves lift bias one and reduced energy bias one-quarter. The radial continuum limit controls discretization; it does not derive the source graph's angular isotropy.

22. A Chapter I field density

Take the original static classical quadrupole field and average its positive free-field energy over angles. Write X² for the sum of the two squared source amplitudes. With kappa=4/5, its total free energy is Uf=X²/R. [1,5]

Conditionally assign Mf=Uf/c² to that free-field account and x=r/R. Its spherical density and enclosed mass are

$$\rho_{\text{in}} = \frac{M_f r^2}{2\pi R^5}, \quad \rho_{\text{out}} = \frac{3M_f R^5}{4\pi r^8},$$

$$\frac{M(<r)}{M_f} = \frac{2}{5}x^5 \quad (x \le 1),$$

$$\frac{M(<r)}{M_f} = 1 - \frac{3}{5}x^{-5} \quad (x \ge 1).$$

Two-fifths of Uf lies inside R and three-fifths outside. Solving the same Poisson receiver, with x=r/R, gives

$$a_{\text{in}} = \frac{2GM_f}{c^2 R}\left(1 - \frac{x^4}{10}\right),$$

$$a_{\text{out}} = \frac{2GM_f}{c^2 r}\left(1 - \frac{x^{-5}}{10}\right).$$

Both a and a' match at R. The source field carries energy outside R; treating that energy as mass makes this region nonvacuum. Its exact correction is proportional to r^-6, rather than a pure vacuum 1/r profile.

At the commuting classical static minimum, each boundary amplitude equals half ε times its source current. Its free energy, interaction and counterterm satisfy

$$U_f + U_{\text{int}} + U_{\text{ct}} = U_f - 2U_f + U_f = 0.$$

This cancellation shows why the positive free-field sector alone is a conditional source choice. The quantum D and C do not commute on the nontrivial lanes; their joint local energy account still needs a derivation. R remains the original coupling radius, distinct from the cube scale and radial outer boundary.

23. Energy unread by the clock

Retain the original sixteen-state source: S is Swap, D=Qleft-Qright, C=iSD, and Eeven=Qleft+Qright+Qleft Qright. Its unnormalized current is Zraw=D+iC. Exact multiplication gives

$$[S, Z_{\mathrm{raw}}] = -2Z_{\mathrm{raw}},$$

$$[E_{\mathrm{even}}, Z_{\mathrm{raw}}] = [D^2 + C^2, Z_{\mathrm{raw}}] = 0.$$

The full Hamiltonian is Hs+Hf+Hint+Hct, with Hs=ε(S+Eeven), Hint=-ε(D phi_plus+C phi_cross), and Hct=$\varepsilon^2\mu(D^2+C^2)$, μ=1/(4R). The free field, both polarizations and both counterterms are retained.

Eeven also commutes with the complete interacting Hamiltonian. Adding any bounded real function F(Eeven) therefore changes energy while leaving the entire current history unchanged for every initial state:

$$Z_{\mathcal{H}_0 + F}(t) = Z_{\mathcal{H}_0}(t).$$

The four diagonal source states have D=C=0 yet distinct energies $\varepsilon(1+qa)^2$. They are current-dark energy controls, not clocks with a nonzero amplitude. Observing the current does not determine all source energy. [5]

A scalar addition kI supplies the simplest common-energy example. Density evolution and relative frequencies stay fixed while the common amplitude phase changes. An external phase reference can compare that phase through its stated coupling and payment; it does not choose a mass baseline by itself.

This distinction also applies to the Si29 account. Independently shifting its positive H0 by kI changes the candidate assigned mass by eu k/c^2. It does not follow from replacing H_{gen} by H_{gen}+kI and taking its absolute value. Reference energy, signed generator, positive account and actual clock receiver must retain their separate definitions.

24. The operational clock response

Supply a physical static-lift perturbation

$$\mathcal{H}(a) = \mathcal{H}_0 + aV + O(a^2).$$

On lane l, let Z=(D+iC)/dl, z=<Z>, and nu=-Im(zdot/z), as in Chapter I. For a bounded V and a fixed preparation, Duhamel gives

$$\delta z(t) = i\int_0^t \langle [V_I(s), Z_I(t)] \rangle \, ds.$$

Where the baseline z0 stays away from zero, the actual phase-rate derivative is

$$\delta\nu = -\mathrm{Im}\left(\frac{\delta\dot{z}}{z_0} - \frac{\dot{z}_0\delta z}{z_0^2}\right).$$

A field-form perturbation needs common energy-form control and a dominated differentiated evolution. An a-dependent preparation supplies an additional state derivative. These requirements belong to the chosen V; the original sixteen-dimensional commutator is only its source algebra.

One exact full-continuum control is the declared coupling of the full interacting Hamiltonian:

$$\mathcal{H}(a) = n(a)\mathcal{H}_0 + f(a)I.$$

It gives

$$z(t; a) = z(n(a)t; 0),$$

$$\nu(t; a) = n(a)\nu(n(a)t; 0).$$

At matched internal age the frequency ratio is n. At equal coordinate time, where nudot exists, its derivative also includes n'(0)t nudot(t;0). The actual retained clock is slightly nonstationary. Accumulated phase follows the time change without requiring an additional second derivative.

Thus a square-root n is an explicit coupling control. A first-order redshift test must specify matched age, a stationary reference, or the transient contribution at equal coordinate durations.

25. Full scaling and ε

The complete scaling control multiplies Hs, Hf, Hint and Hct by the same n. Replacing only ε by n ε at a fixed field instead gives

$$\mathcal{H}_\varepsilon = nH_s + H_f + nH_{\rm int} + n^2 H_{\rm ct}.$$

For s=n'(0), its first-order difference from full scaling is

$$V_\varepsilon - s\mathcal{H}_0 = s(H_{\rm ct} - H_f).$$

For the source-only slope s=-1/2, the residual coefficients on (Hs,Hf,Hint,Hct) are (0,1/2,0,-1/2). This is not a scalar on the full field. The counterterm commutes with the full Hamiltonian and current, so its energy stays unread throughout this history. The free-field part of the residual can change the current.

The retained hard-branch second-order diagnostic is

$$\Omega_l^{(2)} = \Delta(1 - 4g_l^2 J), \qquad \Delta = 2\varepsilon,$$

$$\frac{d\log \Omega_l^{(2)}}{d\log \varepsilon} = 1 - \frac{8g_l^2 J + 4g_l^2 \Delta J'}{1 - 4g_l^2 J}.$$

The hard split is held fixed in this perturbation. J and J' are positive, so the derivative is below one. The six original coupling squares differ, producing six disjoint derivative enclosures. These are second-order source diagnostics, not finite-lift full-field response certificates.

The actual physical V remains the question. A shared perturbation across six separately prepared lanes can test universality; fitting each lane's ε independently would change that test.

The energy unit e_u, action I_u and length $\ell_{\rm phys}$ must be shared between source, clock and field. They set the source time parameter and cube's lift coefficient:

$$\theta_{\rm source} = \frac{e_u t}{I_u}, \qquad \frac{2Ge_u}{c^4 \ell_{\rm phys}}.$$

A physical ε alone does not determine both conversions.

26. Correlated phase and precision

Under the declared full scaling, compare the same baseline phase history at T and nT. Within the common certified window, with $\ell \le \nu_0 \le u$,

$$\delta\theta(T) = -\int_{nT}^{T} \nu_0(s)\, ds,$$

$$-u(1-n)T \le \delta\theta \le -\ell(1-n)T < 0.$$

For n=sqrt(1-a), 0<a<1, rational bounds on 1-n give

$$-\frac{uaT}{2-a} \le \delta\theta \le -\frac{\ell aT}{2}.$$

The signed shift exists for every positive a and T in this window. At the frozen Earth surface lift, all six one-bare-cycle shifts lie between -4.373898 and -4.373888 nanoradians. These are conditional source predictions, not achieved apparatus sensitivity. [5]

A separate comparison encloses the two phases independently. Those bands are disjoint only under the sufficient condition $a > 1-(\ell/u)^2$. Its conservatively rounded thresholds are:

Lane	Sufficient a
01	2.84407×10^{-9}
02	1.22170×10^{-8}
03	1.89316×10^{-6}
12	3.27189×10^{-9}
13	1.74924×10^{-6}
23	1.60120×10^{-6}

These thresholds concern independent enclosures; they are not a minimum lift for the correlated phase sign. A phase error δ on each comparison needs

$$\ell(1-n)T > 2\delta.$$

The physical coupling, common preparation and unwrapped phase branch remain explicit. Chapter II's fresh preparations, timing/error budgets, paid reference and turn counter supply the separate measurement protocol. [2]

27. Frozen anchors and reproduction

The recovered CR004 premises retain the same declared gravitational parameters and radii. Recalculation gives the following spherical surface values; escape speeds are km/s. [4]

Body	Lift a(b)	g, m/s²	Escape
Earth	1.39225×10^{-9}	9.82025	11.1861
Moon	6.28226×10^{-11}	1.62490	2.37618
Sun	4.24501×10^{-6}	274.200	617.675

The formulas use the same frozen μ=GM and radius on both sides. These inputs preserve the intended Newtonian recovery; they do not determine mass from the microscopic account or select G. The digits expose arithmetic, rather than measured precision or corrections for shape and rotation.

The finite cube and shell controls add different information: a source-shape correction, exact source flux, boundary energy and quantified continuum error. The interacting-clock controls add the complete perturbation and a visible precision target.

Reproduction retains editable manuscript, exact operands, source snapshots, native receipts, independent code and content hashes. The source checker has 2,040 exact assertions, the radial checker 312, and the clock checker 822. The static-field checker adds 52 assertions, the kernel/frozen-arithmetic checker 45, the uniform-coupling certificate 23, and the correlated-phase supplement 47. Fresh native execution and custody are recorded separately in the campaign receipt map; retained historical calls are not counted as new work.

The old draft's 42 symbolic checks remain historical. They are not relabeled as the present verification. CR004, G224's retrieved index and LC03 retain their original evidence scopes. The original G224 derivation is still unrecovered; LC03 is an imported mechanism replay, and the current retained source route is the Si29 construction used here.

28. The next physical derivations

First derive the actual local energy/stress coupling, including field, interaction and reference terms. Carry its spatial density through one calibrated receiver.

Next derive a graph or field response with a massless isotropic k^2 leading term and bounded spatial/frequency error, retaining the radial boundary account.

Then apply one physical V to all six clocks, with matched-age or transient timing explicit. Derive causal kinetics and the spatial metric response, including γ, using the same source normalization.

Memory and closure methods control eliminated information. The spatial operator and metric coupling need their own derivations.

References and retained sources

[1] Brady, ChatGPT and Codex. *Volume I, Chapter I*, final draft and sam_volume_I_i source/certificate, 15 October 2026.

[2] Same authors. *Chapter II: Retention, accumulation and the finite ledger*, 20-page physical development draft and this revision's companion source-to-space update.

[3] SAM Research Project. volii_duck_expansion1/2, Si29 effects; native_energy_source1, radial response; current source/radial reports and operands.

[4] SAM Research Project. CR004 declared premises, precommit and scoped weak-field result. Current arithmetic uses the frozen premises.

[5] Current clock/static-field derivations, certificates, checks and native receipt map, companion archive.

[6] SAM Research Project. native_coupling_scale1 and source_spatial_response1: selected coupling and frequency-dependent source-response scopes.

[7] C. M. Will. *The Confrontation between General Relativity and Experiment*. Living Reviews in Relativity 17, 4 (2014), doi:10.12942/lrr-2014-4. Conventional metric notation.